

YEAR 7 MATHEMATICS

Chapter 1

Integers

WORKSHEETS

1

How to use this booklet

- 1** There is **one worksheet for each lesson**. It has three columns that get harder from left to right: **Easy**, then **Medium**, then **Challenging**.

WE DO

Top of each column. Your teacher works the examples on the board. **Copy every step** into the space.

- 2** Read the **speech bubble** at the top of each column: it tells you what to do. A new bubble means a new kind of question.

YOU DO

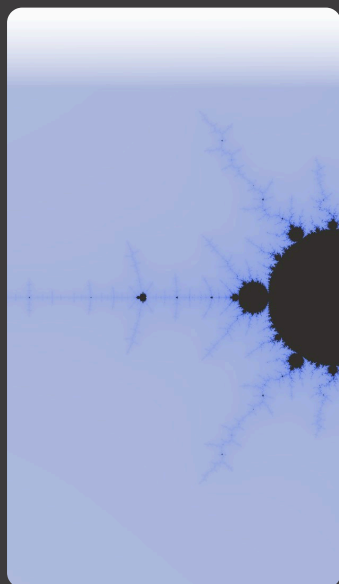
The rest of the column. Your turn. Column 1: answer on the line. Columns 2 and 3: show your working on the lines.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
1.01	Integers and opposites	3		☐
1.02	The number line	5		☐
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1.04	Adding integers	9		☐
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In this chapter you will

- understand positive and negative numbers and their opposites
- use a number line to show and compare integers
- order integers
- add, subtract, multiply and divide integers
- use the order of operations with integers, with and without a calculator
- solve problems involving integers

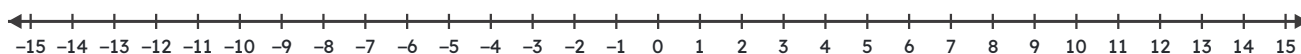


About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the long antenna of the set along the negative real axis, dotted with tiny copies of the whole set. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.



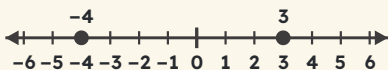
Step 1 What is an integer?

Easy

Questions 1–10

HOW TO

An **integer** is a whole number. It can be positive, negative or zero.
Negative integers are to the **left** of 0.
Positive integers are to the **right**.

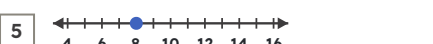
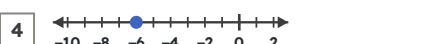
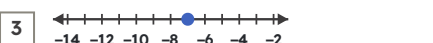
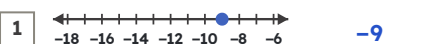


Which integer is marked by the dot? Count along from a number you know.

WE DO With your teacher: copy the working.



YOU DO Your turn.



Write an integer for each situation.

WE DO With your teacher: copy the working.

E3 38 m above sea level _____

E4 39 floors down _____

YOU DO Your turn.

6 a loss of \$23 _____

7 72 floors up _____

8 a loss of \$85 _____

9 a gain of 37 kg _____

10 a rise of 43 cm _____

Step 2 Integers in real life

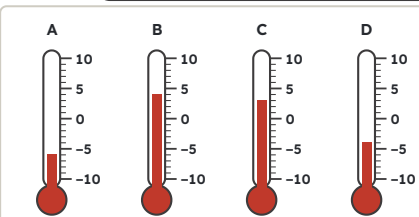
Easy

Questions 11–17

HOW TO

Use a **negative** integer for below zero, down, a loss, a fall or a withdrawal.
Use a **positive** integer for above zero, up, a gain, a rise or a deposit.
Zero is neither: it is the starting point.

Read each thermometer. Write the temperature as an integer.



WE DO With your teacher: copy the working.

E1 Thermometer A _____

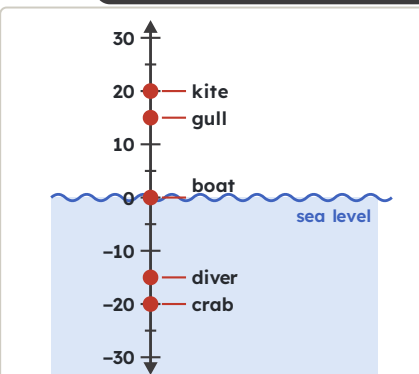
YOU DO Your turn.

11 Thermometer B _____ **4°C**

12 Thermometer C _____

13 Thermometer D _____

Write each height as an integer. Above sea level is positive, below is negative.



WE DO With your teacher: copy the working.

E2 the kite _____

YOU DO Your turn.

14 the gull _____ **15 m**

15 the boat _____

16 the diver _____

17 the crab _____

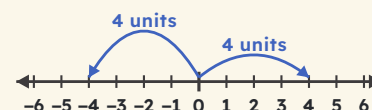
Step 3 Opposites

Medium

Questions 18–21

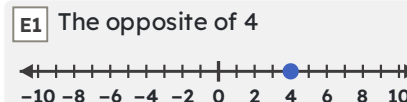
HOW TO

The **opposite** of an integer is the same distance from 0, on the other side.
The opposite of 4 is -4. The opposite of -4 is 4. The opposite of 0 is 0.

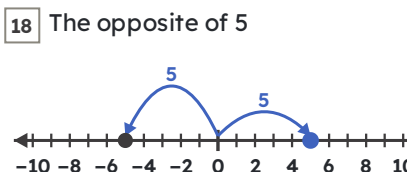


The dot shows an integer. Mark its opposite, then write it. Show the distance from 0 on both sides.

WE DO With your teacher: copy the working.



YOU DO Your turn.



5 is 5 units from 0

opposite: -5

19 The opposite of 7

7 is 7 units from 0

opposite: -7

Write the opposite of each integer.

WE DO With your teacher: copy the working.

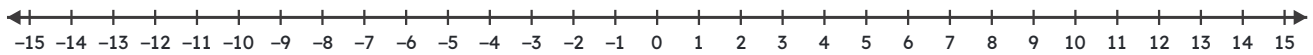
E2 57 _____

E3 128 _____

YOU DO Your turn.

20 -65 _____

21 111 _____



Step 4 The opposite of the opposite

Medium

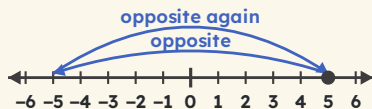
Questions 22–29

HOW TO

A $-$ sign in front of a number means the **opposite** of.

$-(-5)$ means the opposite of -5 , which is 5.

$-(+5)$ means the opposite of 5, which is -5 .



Simplify. Say "the opposite of" for each $-$ in front of a bracket.

WE DO With your teacher: copy the working.

E1 $-(-33)$ E2 $-(+6)$

YOU DO Your turn.

22 $-(-60)$ 23 $-(+23)$

= opp. of -60 = opp. of 23

= 60 = ____

24 $-(-24)$ 25 $-(+27)$

= opp. of ____

= ____

Find $-n$ for each value of n . Put the value in brackets first.

WE DO With your teacher: copy the working.

E3 $n = 15$

YOU DO Your turn.

26 $n = -22$ 27 $n = 20$

$-n = -(-22)$ $-n = -(20)$

= 22 = ____

28 $n = -28$ 29 $n = -30$

$-n = -(\text{____})$

= ____

Step 5 Distance from zero

Challenging

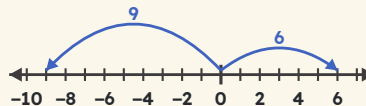
Questions 30–34

HOW TO

The **distance from 0** is how many units a number is from 0. A distance is never negative.

-6 and 6 are both 6 units from 0: they are opposites.

-9 is further from 0 than 6, even though 6 is larger.



Answer each question. Write the distances from 0 on the first line.

WE DO With your teacher: copy the working.

E1 Which is further from 0: -8 or 9?

E2 Which two integers are 8 units from 0?

E3 List every integer less than 4 units from 0.

YOU DO Your turn.

30 Which is further from 0: -7 or 3?

-7 : 7 units, 3: 3 units

further: ____

31 Which two integers are 7 units from 0?

____ right of ____: ____

____ left of ____: ____

32 List every integer less than 3 units from 0.

33 Which is further from 0: -2 or 3?

34 Which two integers are 9 units from 0?

Step 6 Opposites in problems

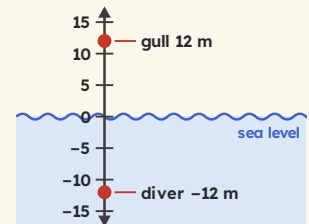
Challenging

Questions 35–37

HOW TO

Write each amount as an integer first. Then use opposites: they are the same distance from 0, on either side.

A gull 12 m above the sea and a diver 12 m below are at opposite heights: 12 and -12 .



Answer each question. Write your reasoning on the lines.

WE DO With your teacher: copy the working.

E1 The opposite of a number is -14 . What is the number?

E2 A diver is 13 m below sea level. A gull is the same distance above. Write both heights as integers.

E3 Two integers are opposites and are 20 units apart. What are they?

YOU DO Your turn.

35 The opposite of a number is -17 . What is the number?

opposite of ? = -17

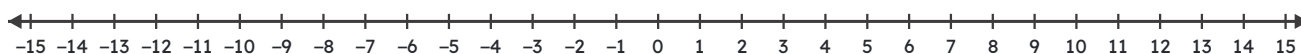
the number is ____

36 A diver is 6 m below sea level. A gull is the same distance above. Write both heights as integers.

diver: ____ m

gull: ____ m

37 Two integers are opposites and are 30 units apart. What are they?



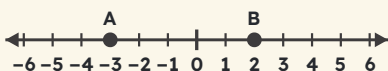
Step 1 Reading a number line

Easy

Questions 1–10

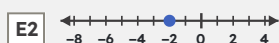
HOW TO

Numbers get larger as you move right, and smaller as you move left. Every mark is one more than the mark to its left. Count along from a number you know.

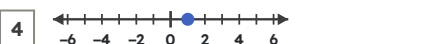
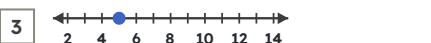
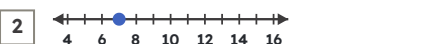
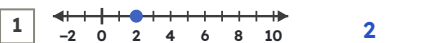


Which integer is marked by the dot?

WE DO With your teacher: copy the working.



YOU DO Your turn.



Which integer is further to the right? Use the number line at the top of the page.

WE DO With your teacher: copy the working.

E3 -9 or 4

E4 10 or -9

YOU DO Your turn.

5 0 or -12

6 0 or 2

7 -5 or -9

8 -8 or -11

9 -5 or -10

10 14 or 12

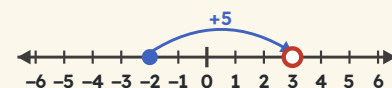
Step 2 Moving along the number line

Easy

Questions 11–14

HOW TO

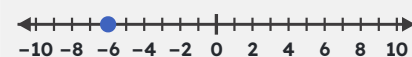
Start at the first number. Moving right makes the number larger; moving left makes it smaller. Start at -2 and move 5 right: you land on 3.



Draw the jump on the number line. Write where you land.

WE DO With your teacher: copy the working.

E1 Start at -6 and move 4 left.



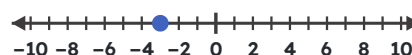
YOU DO Your turn.

11 Start at 0 and move 7 left.



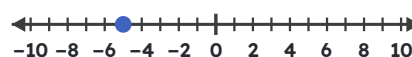
land on -7

12 Start at -3 and move 7 left.



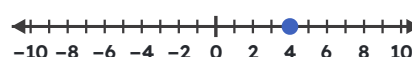
land on ____

13 Start at -5 and move 5 left.



land on ____

14 Start at 4 and move 5 right.



Step 3 Scales that go up in 2s, 5s or 10s

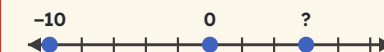
Medium

Questions 15–18

HOW TO

Not every number line goes up in ones. To find the step, take two labelled marks: find the difference, then divide by the number of gaps between them.

From -10 to 0 is 10, in 5 gaps: each gap is 2.



Find the step, then the number at the ?

WE DO With your teacher: copy the working.

E1 What number is at the ?



YOU DO Your turn.

15 What number is at the ?



3 gaps = 30, so step = 10

? = 20

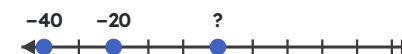
16 What number is at the ?



2 gaps = 10, so step = 5

? = ____

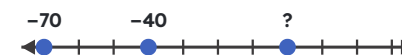
17 What number is at the ?

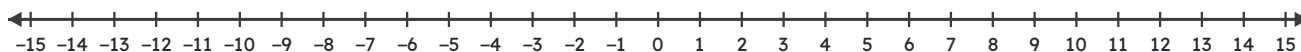


____ gaps = ____, so step = ____

? = ____

18 What number is at the ?





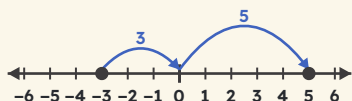
Step 4 Distance between integers

Medium

Questions 19–21

HOW TO

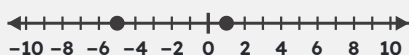
Count the units from one integer to the other. If one is negative and one is positive, count to 0, then on from 0.
-3 to 5: 3 units to 0, then 5 more. $3 + 5 = 8$ units.



How many units apart are the two integers? Draw the jumps through 0.

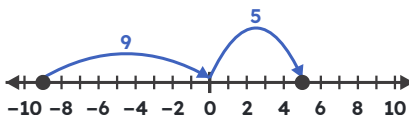
WE DO With your teacher: copy the working.

E1 -5 and 1



YOU DO Your turn.

19 -9 and 5



-9 to 0: 9, 0 to 5: 5

$9 + 5 = 14$

How many units apart? Both are negative: count from one to the other.

WE DO With your teacher: copy the working.

E2 -14 and -2

YOU DO Your turn.

20 -15 and -4

-15 to -4

11 units

21 -9 and -1

-9 to -1

8 units

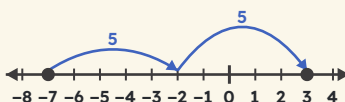
Step 5 Halfway between

Challenging

Questions 22–24

HOW TO

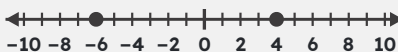
Find the distance between the two integers and halve it. Then move that far from the smaller one.
-7 and 3 are 10 apart. Half is 5. $-7 + 5 = -2$.



Which integer is exactly halfway between? Show the two equal jumps.

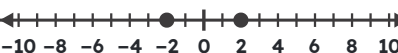
WE DO With your teacher: copy the working.

E1 -6 and 4



YOU DO Your turn.

22 -2 and 2



4 apart, half is 2

___ + ___ = ___

Now without a number line.

WE DO With your teacher: copy the working.

E2 Halfway between -9 and 7

YOU DO Your turn.

23 Halfway between -24 and -4

20 apart, half is 10

___ + ___ = ___

24 Halfway between -21 and -7

___ apart, half is ___

___ + ___ = ___

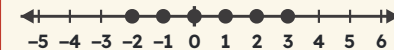
Step 6 Number line problems

Challenging

Questions 25–26

HOW TO

Draw a quick number line. Mark the numbers you know first.
Between does not include the end numbers: the integers between -3 and 4 are -2, -1, 0, 1, 2 and 3 (six of them).



Answer each question. Show your working.

WE DO With your teacher: copy the working.

E1 How many integers are between -9 and 11?

E2 An ant starts at 6. It walks 12 units left, then 7 units right. Where is it now?

E3 Point A is at -10. Point B is 8 units to the right of A. Where is B, and how far is B from 0?

YOU DO Your turn.

25 How many integers are between -17 and 9?

-16 to -1: 16, and 0

1 to 8: 8

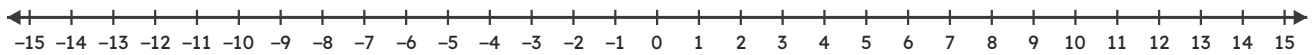
___ + ___ + ___ = ___

26 An ant starts at 5. It walks 8 units left, then 2 units right. Where is it now?

___ - ___ = ___

___ + ___ = ___

it is at ___



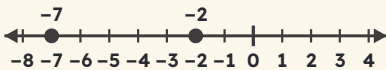
Step 1 Which is larger?

Easy

Questions 1–9

HOW TO

On a number line, the integer further to the right is larger.
 -2 is to the right of -7 , so -2 is larger than -7 .
 Every positive integer is larger than every negative integer.



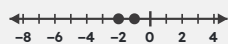
Which integer is larger?
 Look at where they are on the number line.

WE DO With your teacher: copy the working.

E1 -4 or -2

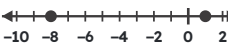


E2 -1 or -2

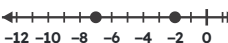


YOU DO Your turn.

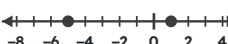
1 1 or -8



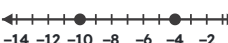
2 -2 or -7



3 -5 or 1



4 -10 or -4



Write the smaller integer.

WE DO With your teacher: copy the working.

E3 3 or 6

E4 -19 or -30

YOU DO Your turn.

5 -4 or -9

6 -6 or -17

7 -13 or 4

8 -17 or 5

9 5 or 4

Step 2 Using < and >

Easy

Questions 10–22

HOW TO

$<$ means "is less than". $>$ means "is greater than".
 The open side of the symbol faces the larger number.
 $-5 < 2$ (-5 is less than 2)
 $3 > -8$ (3 is greater than -8)
 $-1 > -6$ (-1 is further right)

Write $<$ or $>$ in the box.

WE DO With your teacher: copy the working.

E1 -12 0

E2 -19 10

YOU DO Your turn.

10 2 12

11 -14 -20

12 2 1

13 -16 -2

14 -9 -16

15 4 10

16 3 4

17 5 -16

True or false?

WE DO With your teacher: copy the working.

E3 $-7 > -4$

E4 $3 > 6$

YOU DO Your turn.

18 $1 > -10$

19 $-15 < 1$

20 $-3 < 6$

21 $0 > -14$

22 $-7 < -10$

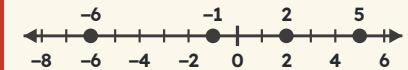
Step 3 Ordering integers

Medium

Questions 23–26

HOW TO

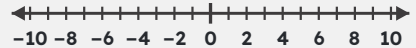
Ascending order: smallest to largest, the order they sit on the number line from left to right.
Descending order: largest to smallest.
 Mark the numbers on a number line, then read them off.



Mark the integers on the number line, then write them in ascending order.

WE DO With your teacher: copy the working.

E1 $9, 2, -1, -6$



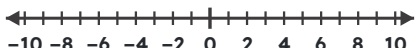
YOU DO Your turn.

23 $-2, 1, -4, 2$



$-4, -2, 1, 2$

24 $-2, 3, -4, -5$



Write the integers in descending order (largest first).

WE DO With your teacher: copy the working.

E2 $22, -24, -7, -19, -31$

YOU DO Your turn.

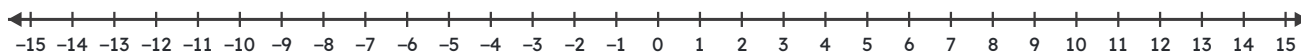
25 $-28, -25, 25, -15, 3$

largest: 25, smallest: -28

$25, 3, -15, -25, -28$

26 $-3, -39, 6, -31, -8$

largest: 6, smallest: -39



Step 4 Comparing temperatures

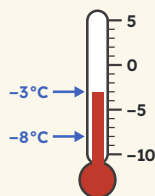
Medium

Questions 27–31

HOW TO

Colder temperatures are lower on a thermometer, so they are smaller numbers.

-8°C is colder than -3°C , so $-8 < -3$.



The table shows the lowest temperature in five cities on one winter day. Use it to answer the questions.

City	Lowest temperature
Beijing	7°C
Toronto	0°C
Hobart	5°C
Denver	-4°C
Oslo	-6°C

WE DO With your teacher: copy the working.

E1 Which city was the coldest?

YOU DO Your turn.

27 Which city was the warmest?

highest: 7°C

Beijing

28 How many cities had a temperature below zero?

below 0: $-4, -6$

___ cities

29 Write the temperatures in ascending order.

___ $^{\circ}\text{C}$, ___ $^{\circ}\text{C}$, ___ $^{\circ}\text{C}$, ___ $^{\circ}\text{C}$, ___ $^{\circ}\text{C}$

30 Which cities were colder than 1°C ?

31 Write $<$ or $>$: Toronto ☐ Denver

Step 5 Simplify, then compare

Challenging

Questions 32–37

HOW TO

Simplify any $-(-n)$ first: $-(-6) = 6$.

Then place the numbers on a number line and compare them.

$-(-2)$, -5 , 0 : simplify to 2 , -5 , 0 .

Ascending: -5 , 0 , 2 .

Simplify first, then answer the question.

WE DO With your teacher: copy the working.

E1 Which is the smallest: $-(-2)$, -9 or $-(-12)$?

E2 Write in ascending order: $-(-4)$, -8 , -9 , 0

YOU DO Your turn.

32 Which is the smallest: $-(-6)$, -3 or $-(-8)$?

$-(-6) = 6$, $-(-8) = 8$

smallest: ___

33 Write in ascending order: $-(-11)$, -3 , -2 , 0

$-(-11) = 11$

___, ___, ___, ___

34 Which is the smallest: $-(-2)$, -6 or $-(-10)$?

35 Write in ascending order: $-(-9)$, -10 , -2 , 0

36 Which is the smallest: $-(-4)$, -6 or $-(-3)$?

37 Write in ascending order: $-(-10)$, -12 , -4 , 0

Step 6 Between, above and below

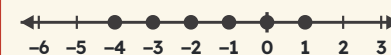
Challenging

Questions 38–41

HOW TO

"Greater than -5 " means to the right of -5 . "Less than 2 " means to the left of 2 .

The integers greater than -5 and less than 2 are -4 , -3 , -2 , -1 , 0 and 1 .



Answer each question. Draw a quick number line if it helps.

WE DO With your teacher: copy the working.

E1 Write three integers greater than -20 but less than -13 .

E2 How many integers are greater than -8 and less than 8 ?

E3 I am an odd integer. I am less than 0 but greater than -8 . What could I be?

YOU DO Your turn.

38 Write three integers greater than -14 but less than -6 .

integers from -13 to -7

e.g. ___, ___, ___

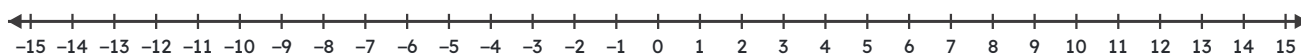
39 How many integers are greater than -6 and less than 10 ?

___ up to ___

___ + ___ + ___ = ___

40 I am an odd integer. I am less than 0 but greater than -9 . What could I be?

41 Write three integers greater than -19 but less than -11 .



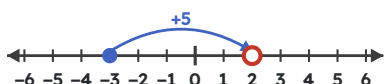
Step 1 Adding a positive Easy

HOW TO

Start at the first number.

Adding a positive number: move right.

$-3 + 5$: start at -3 , move 5 right.
Land on 2.



Draw the jump. Write where you land.

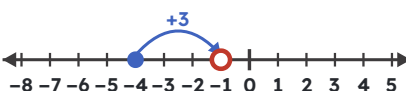
WE DO

E1 $-2 + 4$



YOU DO

1 $-4 + 3$



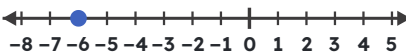
$= -1$

2 $-5 + 2$



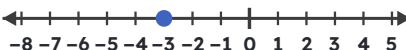
$=$ ____

3 $-6 + 4$

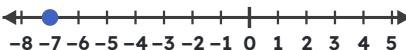


$=$ ____

4 $-3 + 7$



5 $-7 + 7$



Step 2 Adding a negative Easy

HOW TO

Adding a negative number: move left.

$+(-4)$ means the same as -4 .

$3 + (-5) = 3 - 5 = -2$



Rewrite $+(-)$ as $-$. Then work it out.

WE DO

E2 $6 + (-2)$

E3 $-2 + (-3)$

YOU DO

6 $7 + (-3)$

7 $5 + (-5)$

$= 7 - 3$

$= 5 - 5$

$= 4$

$=$ ____

8 $2 + (-6)$

9 $4 + (-10)$

$=$ ____ $-$ ____

$=$ ____

10 $-1 + (-4)$

11 $-3 + (-6)$

Step 3 Zero pairs Medium

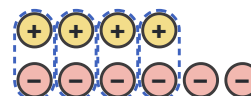
HOW TO

A $+$ and a $-$ make a zero pair.

They cancel out.

So take the difference. The sign with more wins.

$-6 + 4$: $6 - 4 = 2$. $-$ has more: -2 .



Take the difference. Which sign has more?

WE DO

E4 $-5 + 8$

E5 $6 + (-9)$

YOU DO

12 $-7 + 10$

13 $-9 + 4$

$10 - 7 = 3$

$9 - 4 = 5$

$+$ has more: 3

$-$ has more: ____

14 $8 + (-12)$

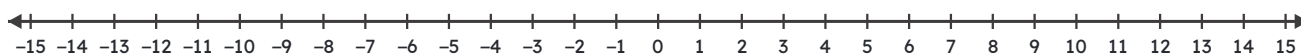
15 $-15 + 20$

____ $-$ ____ $=$ ____

$-$ has more: ____

16 $25 + (-40)$

17 $-36 + 50$



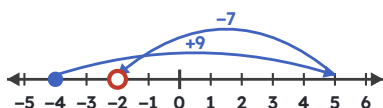
Step 4 Three numbers

Medium

HOW TO

Add from left to right, one jump at a time.

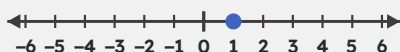
$$-4 + 9 + (-7) = 5 + (-7) = -2$$



Add from left to right.
Draw the jumps when
there is a number line.

WE DO

E6 $1 + (-4) + 5$



YOU DO

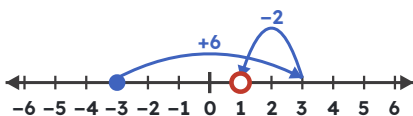
18 $2 + (-5) + 4$



$$= -3 + 4$$

$$= 1$$

19 $-3 + 6 + (-2)$



$$= 3 + (-2)$$

$$= 1$$

20 $12 + (-15) + (-6)$

$$= 12 + (-21)$$

$$= -9$$

21 $-20 + 8 + 15$

Step 5 Adding in real life

Challenging

HOW TO

Write the story as a sum.

Up, rises and gains are **positive**.

Down, falls and spending are **negative**.

-6°C , then it warms up 10°C :

$$-6 + 10 = 4^{\circ}\text{C}$$

Write a sum. Then answer
the question.

WE DO

E7 It is -5°C . The temperature
rises 8°C . What is the
temperature now?

YOU DO

22 It is -4°C at dawn. By lunch
it has warmed up 9°C . What
is the temperature at lunch?

$$-4 + 9$$

$$= 5^{\circ}\text{C}$$

23 A lift is at level -2 . It goes
up 5 floors. Which level is it
on now?

$$-2 + 5$$

$$= \text{level } 3$$

24 Sam has \$15 in the bank.
Sam spends \$40 using
credit. What is the balance
now?

$$15 + (-40)$$

$$= -25$$

25 A diver is at -18 m . She
swims up 7 m , then down
 4 m . Where is she now?

Step 6 Is it right?

Challenging

HOW TO

Work it out yourself first.

Then compare. Say yes or no, and
give the right answer.

Kim says $3 + (-5) = -2$.

$3 - 5 = -2$. Yes, Kim is right.

Is it right? Explain your
answer.

WE DO

E8 Tom says $-6 + 2 = -8$. Is he
right?

YOU DO

26 Mia says $-8 + 5 = -13$. Is she
right?

start at -8 , move 5 right

$$= -3$$

No. The answer is -3 .

27 Ben says $-7 + 3 + (-3) = -7$.
Is he right?

$3 + (-3)$ is a zero pair

$$-7 + 0 = -7$$

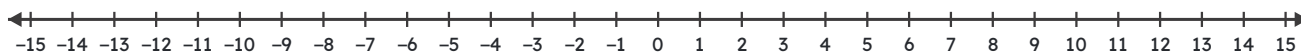
28 Ali says $-4 + \square = 2$ means $\square$
 $= -2$. Is he right?

from -4 to 2 is 6 right

$$\square = 6$$

No. $\square$ is 6 , not -2 .

29 Jo says adding always
makes a number bigger. Is
Jo right?



Step 1 Subtracting a positive: move left

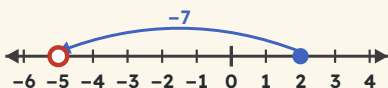
Easy

Questions 1–6

HOW TO

Subtracting a positive number means move left.

2 – 7: start at 2, move 7 left, land on –5.



Draw the jump left, then write the answer.

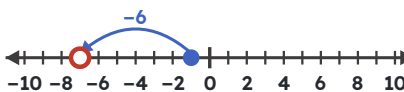
WE DO With your teacher: copy the working.

E1 0 – 4



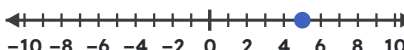
YOU DO Your turn.

1 –1 – 6



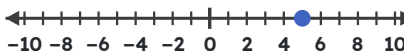
= –7

2 5 – 10



= ____

3 5 – 9



= ____

Evaluate. Picture the jump left.

WE DO With your teacher: copy the working.

E2 11 – 12

E3 0 – 15

YOU DO Your turn.

4 2 – 8

–6

5 9 – 12

6 6 – 10

Step 2 Starting below zero

Easy

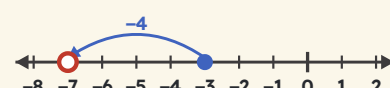
Questions 7–14

HOW TO

Starting below zero and subtracting takes you further below zero.

$$-3 - 4 = -7$$

Think of money: you owe \$3, then spend \$4 more. Now you owe \$7.



Draw the jump, then write the answer.

WE DO With your teacher: copy the working.

E1 –3 – 5



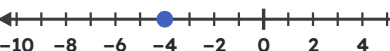
YOU DO Your turn.

7 –4 – 2



= –6

8 –4 – 5



= ____

Evaluate.

WE DO With your teacher: copy the working.

E2 –3 – 10

E3 –17 – 12

YOU DO Your turn.

9 –12 – 16

–28

10 –9 – 19

11 –1 – 14

12 –3 – 16

13 –8 – 18

14 –19 – 17

Step 3 Subtracting a negative: move right

Medium

Questions 15–18

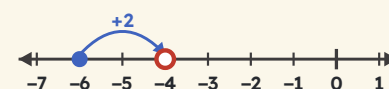
HOW TO

Subtracting a negative is the same as adding its opposite: – (–3) becomes + 3.

$$5 - (-3) = 5 + 3 = 8$$

$$-6 - (-2) = -6 + 2 = -4$$

Taking away a debt of \$3 leaves you \$3 better off.



Change – (–) to +. Draw the jump right, then write the answer.

WE DO With your teacher: copy the working.

E1 –2 – (–6)



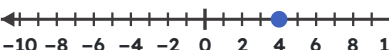
YOU DO Your turn.

15 –7 – (–4)



= –7 + 4 = –3

16 4 – (–6)



= ____ + ____ = ____

Change – (–) to +, then evaluate.

WE DO With your teacher: copy the working.

E2 –5 – (–7)

YOU DO Your turn.

17 10 – (–7)

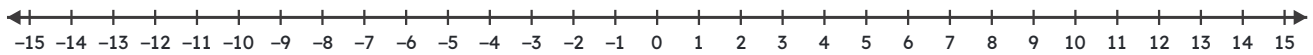
= 10 + 7

= 17

18 –6 – (–2)

= –6 + 2

= ____

**Step 4 Temperature change****Medium**

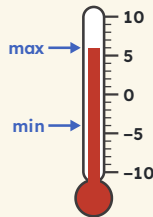
Questions 19–19

HOW TO

Change in temperature = new – old. A positive change is a rise; a negative change is a fall.

From -4°C to 6°C : $6 - (-4) = 6 + 4 = 10$, a rise of 10°C .

On the thermometer, count the degrees from one reading to the other.



Find the difference between the maximum and minimum temperatures.

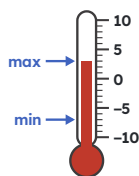
WE DO With your teacher: copy the working.

E1 Minimum -9°C , maximum 9°C



YOU DO Your turn.

19 Minimum -7°C , maximum 3°C



$$3 - (-7)$$

$$= 3 + 7$$

$$= 10^{\circ}\text{C}$$

Find the change: new – old. Say whether it is a rise or a fall.

WE DO With your teacher: copy the working.

E2 From 11°C to -9°C

YOU DO Your turn.

Step 5 Mixed + and –**Challenging**

Questions 20–24

HOW TO

First change every $-(-n)$ to $+n$. Then work from left to right, one step at a time.

$$\begin{aligned} -3 - (-5) + (-4) \\ = -3 + 5 + (-4) \\ = 2 + (-4) \\ = -2 \end{aligned}$$

Evaluate. Change $-(-)$ to $+$ first, then add from left to right.

WE DO With your teacher: copy the working.

E1 $5 - (-11) + (-6)$

YOU DO Your turn.

20 $12 - (-10) + (-9)$

$$= 12 + 10 + (-9)$$

$$= 22 + (-9)$$

$$= \underline{\hspace{2cm}}$$

21 $-11 - (-11) + (-7)$

$$= \underline{\hspace{2cm}} + \underline{\hspace{2cm}} + (\underline{\hspace{2cm}})$$

$$= \underline{\hspace{2cm}} + (\underline{\hspace{2cm}})$$

$$= \underline{\hspace{2cm}}$$

22 $-12 - (-2) + (-4)$

23 $-8 - (-13) + (-2)$

24 $-8 - (-9) + (-5)$

Step 6 Heights and depths**Challenging**

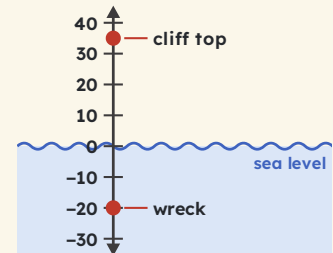
Questions 25–27

HOW TO

Heights above sea level are positive; depths below are negative.

Difference in height = higher – lower.

From a wreck at -20 m to a cliff top at 35 m : $35 - (-20) = 55\text{ m}$.



Write a number sentence, then answer the question.

WE DO With your teacher: copy the working.

E1 A submarine at -99 m rises 32 m . How deep is it now?

E2 A bird is 87 m above sea level and a fish is 21 m below. How far apart are they?

E3 What is 8 less than -12 ?

YOU DO Your turn.

25 A submarine at -124 m rises 18 m . How deep is it now?

$$-124 + 18$$

$$= \underline{\hspace{2cm}}\text{ m}$$

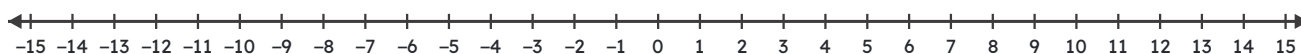
26 A bird is 53 m above sea level and a fish is 40 m below. How far apart are they?

$$\underline{\hspace{2cm}} - (\underline{\hspace{2cm}})$$

$$= \underline{\hspace{2cm}} + \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}\text{ m}$$

27 What is 12 less than -7 ?



Step 1 Multiplying: equal groups

Easy

Questions 1–8

HOW TO

$3 \times (-2)$ means 3 groups of -2 .

$(-2) + (-2) + (-2) = -6$, so $3 \times (-2) = -6$.



Write the groups as a sum, then as a product.

WE DO With your teacher: copy the working.

E1 $2 \times (-3)$



YOU DO Your turn.

1 $4 \times (-3)$



$(-3) + (-3) + (-3) + (-3)$

$= -12$

2 $4 \times (-2)$



$(-2) + (-2) + (-2) + (-2)$

$=$ _____

3 $2 \times (-2)$



$(\quad) + (\quad)$

$=$ _____

Evaluate. A positive times a negative is negative.

WE DO With your teacher: copy the working.

E2 $5 \times (-7)$

E3 $3 \times (-5)$

YOU DO Your turn.

4 -8×3

-24

5 $2 \times (-4)$

6 $8 \times (-4)$

7 -6×5

8 $6 \times (-9)$

Step 2 The sign rules

Easy

Questions 9–18

HOW TO

Same signs: the answer is **positive**.

Different signs: the answer is **negative**.

Multiply the numbers, then use the table to choose the sign.

$\times$	$+$	$-$
$+$	$+$	$-$
$-$	$-$	$+$

Will the answer be positive or negative? Write $+$ or $-$.

WE DO With your teacher: copy the working.

E1 $4 \times (-6)$

E2 -8×2

YOU DO Your turn.

9 $5 \times (-7)$

$-$

10 8×5

11 8×7

12 $4 \times (-9)$

Now evaluate each product.

WE DO With your teacher: copy the working.

E3 $-6 \times (-8)$

E4 -4×3

YOU DO Your turn.

13 $-7 \times (-2)$

14

14 $-3 \times (-7)$

15 $9 \times (-3)$

16 $-6 \times (-10)$

17 $7 \times (-10)$

18 $-8 \times (-3)$

Step 3 Why $- \times -$ is $+$

Medium

Questions 19–26

HOW TO

Look down the pattern. Each time the first number goes down by 1, the answer goes up by 3.

So the pattern carries on into positive answers.

product	answer	
$2 \times (-3)$	-6	
$1 \times (-3)$	-3	$+3$
$0 \times (-3)$	0	$+3$
$-1 \times (-3)$	3	$+3$
$-2 \times (-3)$	6	$+3$

Continue the pattern for three more lines.

WE DO With your teacher: copy the working.

E1 $2 \times (-9) = -18$, $1 \times (-9) = -9$, ...

YOU DO Your turn.

19 $2 \times (-5) = -10$, $1 \times (-5) = -5$, ...

$0 \times (-5) = 0$

$-1 \times (-5) = 5$

$-2 \times (-5) = 10$

20 $2 \times (-7) = -14$, $1 \times (-7) = -7$, ...

$0 \times (-7) = 0$

$-1 \times (-7) = 7$

$___ \times (___) = ___$

Evaluate. Negative times negative is positive.

WE DO With your teacher: copy the working.

E2 $-8 \times (-7)$

E3 $-10 \times (-6)$

YOU DO Your turn.

21 $-7 \times (-8)$

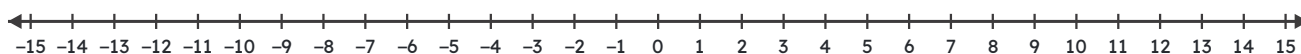
22 $-4 \times (-10)$

23 $-8 \times (-11)$

24 $-11 \times (-12)$

25 $-9 \times (-8)$

26 $-6 \times (-7)$



Step 4 Multiplying three integers

Medium

Questions 27–34

HOW TO

Multiply two numbers at a time, from left to right.

$$\begin{aligned} -2 \times 3 \times (-4) \\ = -6 \times (-4) \\ = 24 \end{aligned}$$

Check: an even number of negatives gives a positive answer; an odd number gives a negative answer.

Evaluate. Multiply two numbers at a time.

WE DO With your teacher: copy the working.

E1 $5 \times (-6) \times 2$

YOU DO Your turn.

27 $-6 \times 5 \times (-3)$

$= -30 \times (-3)$

$= 90$

28 $6 \times 3 \times (-4)$

$= 18 \times (-4)$

$= \underline{\hspace{2cm}}$

29 $2 \times (-5) \times 4$

$= \underline{\hspace{2cm}} \times \underline{\hspace{2cm}}$

$= \underline{\hspace{2cm}}$

30 $2 \times (-5) \times (-3)$

Count the negatives. Is the answer positive or negative?

WE DO With your teacher: copy the working.

E2 $(-7) \times (-2) \times (-9)$

E3 $3 \times 9 \times 2$

YOU DO Your turn.

31 $7 \times (-8) \times 5 \times (-4)$

32 $(-6) \times 2 \times (-8)$

33 $(-3) \times (-2) \times 2$

34 $3 \times 3 \times (-9)$

Step 5 Multiplying in real life

Challenging

Questions 35–37

HOW TO

A change that repeats is a multiplication. Going down or falling is negative.

A diver goes down 3 m every minute for 5 minutes: $-3 \times 5 = -15$. She is 15 m below where she started.



Write a number sentence, then answer the question.

WE DO With your teacher: copy the working.

E1 A diver goes down 3 m every minute. How deep is she after 9 minutes?

E2 The temperature drops 5°C each hour for 6 hours. What is the total change?

E3 A share price falls \$5 a week for 3 weeks. What is the total change?

YOU DO Your turn.

35 A diver goes down 2 m every minute. How deep is she after 9 minutes?

-2×9

$= \underline{\hspace{2cm}} \text{ m}$

36 The temperature drops 2°C each hour for 5 hours. What is the total change?

$\underline{\hspace{2cm}} \times \underline{\hspace{2cm}}$

$= \underline{\hspace{2cm}} ^\circ\text{C}$

37 A share price falls \$3 a week for 10 weeks. What is the total change?

Step 6 Missing numbers and squares

Challenging

Questions 38–41

HOW TO

Find a missing number by dividing: $-4 \times \square = 28$, so $\square = 28 \div (-4) = -7$.

Squaring: $(-5)^2 = -5 \times (-5) = 25$. The square of any integer is never negative.

Answer each question. Show your working.

WE DO With your teacher: copy the working.

E1 Find the missing number: $-3 \times \square = -15$

E2 Evaluate $(-7)^2$.

E3 Find two integers with a product of -35 and a sum of 2.

YOU DO Your turn.

38 Find the missing number: $-8 \times \square = 48$

$\square = 48 \div (-8)$

$\square = \underline{\hspace{2cm}}$

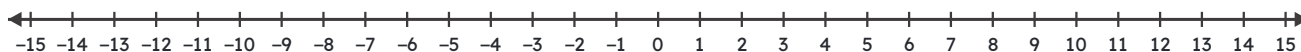
39 Evaluate $(-4)^2$.

$= \underline{\hspace{2cm}} \times \underline{\hspace{2cm}}$

$= \underline{\hspace{2cm}}$

40 Find two integers with a product of -15 and a sum of 2.

41 Find the missing number: $-7 \times \square = -21$



Step 1 Division undoes multiplication

Easy

Questions 1–10

HOW TO

Every multiplication fact gives two division facts.

$$3 \times (-4) = -12, \text{ so}$$

$$-12 \div 3 = -4 \text{ and } -12 \div (-4) = 3.$$

-12 shared into 3 equal groups is -4 in each group.



Write two division facts from the multiplication fact.

WE DO With your teacher: copy the working.

E1 $2 \times (-6) = -12$

YOU DO Your turn.

1 $4 \times (-7) = -28$

2 $3 \times (-5) = -15$

$-28 \div 4 = -7$

$-15 \div 3 = -5$

$-28 \div (-7) = 4$

$___ \div (___) = ___$

3 $3 \times (-2) = -6$

4 $2 \times (-7) = -14$

$___ \div ___ = ___$

$___ \div (___) = ___$

Evaluate. Share the negative number into equal groups.

WE DO With your teacher: copy the working.

E2 $-60 \div 6$

E3 $-32 \div 4$

YOU DO Your turn.

5 $-45 \div 5$

-9

6 $-20 \div 5$

7 $-24 \div 8$

8 $-81 \div 9$

9 $-6 \div 3$

10 $-80 \div 10$

Step 2 The sign rules for dividing

Easy

Questions 11–20

HOW TO

The rules are the same as for multiplying.

Same signs: positive. Different signs: negative.

$$-20 \div (-4) = 5$$

$$20 \div (-4) = -5$$

$\div$	+	-
+	+	-
-	-	+

Evaluate each quotient. Decide the sign first.

WE DO With your teacher: copy the working.

E1 $-10 \div (-5)$

E2 $-14 \div (-7)$

YOU DO Your turn.

11 $-32 \div (-4)$

8

12 $-90 \div 9$

13 $-20 \div (-4)$

14 $-18 \div (-6)$

15 $-12 \div (-6)$

16 $-36 \div (-4)$

17 $-100 \div 10$

Divide, then check your answer by multiplying.

WE DO With your teacher: copy the working.

E3 $63 \div (-7)$

YOU DO Your turn.

18 $-56 \div (-8)$

$= 7$

check: $7 \times (-8) = -56$

19 $-35 \div (-7)$

$= 5$

check: $___ \times (___) = ___$

20 $10 \div (-5)$

$= ___$

check: $___ \times (___) = ___$

Step 3 Missing numbers

Medium

Questions 21–28

HOW TO

Use the opposite operation.

$$\square \div (-3) = 5$$

$$\square = 5 \times (-3) = -15$$

$$-24 \div \square = 6$$

$$\square = -24 \div 6 = -4$$

Find the missing number. Use the opposite operation.

WE DO With your teacher: copy the working.

E1 $\square \div (-6) = -3$

E2 $14 \div \square = -2$

YOU DO Your turn.

21 $\square \div 7 = 8$

22 $-72 \div \square = 9$

$\square = 8 \times 7$

$\square = -72 \div 9$

$\square = 56$

$\square = ___$

23 $\square \div (-7) = 5$

24 $-45 \div \square = -9$

$\square = ___ \times (___)$

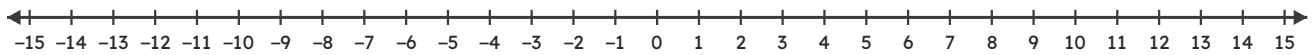
$\square = ___$

25 $\square \div 5 = -3$

26 $27 \div \square = -3$

27 $\square \div (-7) = -7$

28 $28 \div \square = 7$



Step 4 Dividing step by step

Medium

Questions 29–35

HOW TO

Divide from left to right, one step at a time.

$$-60 \div 5 \div (-3)$$

$$= -12 \div (-3)$$

$$= 4$$

A fraction line also means divide: $\frac{-18}{6}$

$$= -18 \div 6 = -3.$$

Evaluate. Divide from left to right.

WE DO With your teacher: copy the working.

E1 $-20 \div 5 \div (-2)$

YOU DO Your turn.

29 $-24 \div 3 \div 2$

$$= -8 \div 2$$

$$= -4$$

30 $-100 \div (-5) \div (-4)$

$$= 20 \div (-4)$$

$$= \underline{\hspace{2cm}}$$

31 $-24 \div 3 \div (-2)$

$$= \underline{\hspace{2cm}} \div (\underline{\hspace{2cm}})$$

$$= \underline{\hspace{2cm}}$$

Write each fraction as a division, then evaluate.

WE DO With your teacher: copy the working.

E2 $\frac{35}{-5}$

YOU DO Your turn.

32 $\frac{-25}{5}$

$$= -25 \div 5$$

$$= -5$$

33 $\frac{-40}{5}$

$$= -40 \div 5$$

$$= \underline{\hspace{2cm}}$$

34 $\frac{-48}{8}$

$$= \underline{\hspace{2cm}} \div \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

35 $\frac{-49}{7}$

$$= \underline{\hspace{2cm}} \div \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

Step 5 Sharing and averages

Challenging

Questions 36–37

HOW TO

Sharing a debt, or a fall, equally is dividing a negative number.

A fall of 12°C over 4 hours: $-12 \div 4 =$

-3 , a fall of 3°C each hour.

The **average** (mean) = total $\div$ how many.

Write a number sentence, then answer the question.

WE DO With your teacher: copy the working.

E1 A debt of \$27 is shared equally by 3 people. Write each share as an integer.

E2 The temperature fell 25°C over 5 hours. What was the change each hour?

E3 Find the average of these temperatures: -3°C , 1°C , -3°C , 1°C .

YOU DO Your turn.

36 A debt of \$150 is shared equally by 5 people. Write each share as an integer.

$$-150 \div 5$$

$$= -30$$

each owes $\underline{\hspace{2cm}}$: $\underline{\hspace{2cm}}$

37 The temperature fell 12°C over 2 hours. What was the change each hour?

$$\underline{\hspace{2cm}} \div \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

$\underline{\hspace{2cm}}^{\circ}\text{C}$ each hour

Step 6 Mixed $\times$ and $\div$

Challenging

Questions 38–43

HOW TO

$\times$ and $\div$ have the same rank: work from left to right.

$$-8 \times 3 \div (-6)$$

$$= -24 \div (-6)$$

$$= 4$$

$$36 \div (-4) \times 2$$

$$= -9 \times 2$$

$$= -18$$

Evaluate. Work from left to right.

WE DO With your teacher: copy the working.

E1 $-3 \times (-6) \div (-3)$

E2 $35 \div (-5) \times 3$

YOU DO Your turn.

38 $-8 \times 3 \div 2$

$$= -24 \div 2$$

$$= \underline{\hspace{2cm}}$$

39 $-12 \div 3 \times 3$

$$= \underline{\hspace{2cm}} \times \underline{\hspace{2cm}}$$

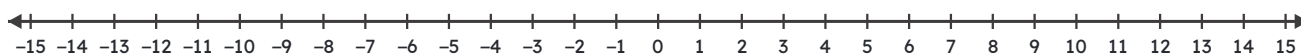
$$= \underline{\hspace{2cm}}$$

40 $4 \times (-4) \div 2$

41 $-21 \div 3 \times (-3)$

42 $4 \times 6 \div (-2)$

43 $12 \div (-6) \times 2$



Step 1 The order of operations

Easy

Questions 1–9

HOW TO

1. Brackets first.
 2. $\times$ and $\div$, from left to right.
 3. $+$ and $-$, from left to right.
- Underline the part you work out first.
- $$-3 + 4 \times (-2) = -3 + (-8) = -11$$

Which operation do you do first? Write it on the line.

WE DO With your teacher: copy the working.

E1 $3 + (-9) \times (-9)$

E2 $-7 \times 8 - (-9)$

YOU DO Your turn.

+ in
brackets

1 $(-4 + (-9)) \times (-5)$

2 $(-9 + (-3)) \times (-3)$

3 $-4 \times (-7) - (-8)$

4 $7 - 9 \div 6$

Underline the first step, then evaluate.

WE DO With your teacher: copy the working.

E3 $2 - (-6) \times (-4)$

YOU DO Your turn.

5 $10 - 4 \times (-3)$

$= 10 - (-12)$

$= 22$

6 $-5 + (-3) \times 6$

$= -5 + (-18)$

$=$

7 $-4 + (-4) \times (-3)$

$=$

$=$

8 $10 + (-6) \times (-5)$

9 $-5 - (-3) \times (-3)$

Step 2 Brackets first

Easy

Questions 10–16

HOW TO

- Work out the brackets first, then the rest.
- $$(-3 + 7) \times 2 = 4 \times 2 = 8$$
- Without brackets the answer changes:
- $$-3 + 7 \times 2 = -3 + 14 = 11$$

Evaluate. Brackets first.

WE DO With your teacher: copy the working.

E1 $(-2 - (-4)) \times 2$

YOU DO Your turn.

10 $(8 - 1) \times (-5)$

$= 7 \times (-5)$

$= -35$

11 $(-9 + 3) \times (-2)$

$= (-6) \times (-2)$

$=$

12 $(-4 - 9) \times (-5)$

$=$

$=$

13 $(2 - 4) \times (-2)$

14 $(-4 - 2) \times 2$

15 $(-4 + (-2)) \times (-4)$

16 $(-4 + (-6)) \times (-3)$

Step 3 $\times$ and $\div$ before $+$ and $-$

Medium

Questions 17–22

HOW TO

- Find every $\times$ and $\div$ and do them first. Then do $+$ and $-$ from left to right.
- $$-10 \div 2 + 3 \times (-4)$$
- $$= -5 + (-12)$$
- $$= -17$$

Evaluate. Do both $\times$ and $\div$ in the first line.

WE DO With your teacher: copy the working.

E1 $-5 \times 3 + 16 \div (-4)$

YOU DO Your turn.

17 $6 \times (-4) - 20 \div 4$

$= -24 - 5$

$= -29$

18 $2 \times 6 + (-12) \div (-4)$

$= 12 + 3$

$=$

19 $-2 \times 4 - (-6) \div (-2)$

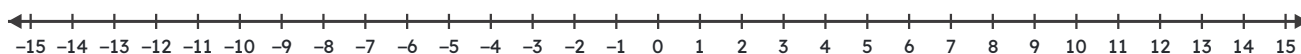
$=$

$=$

20 $4 \times 2 - (-10) \div (-5)$

21 $3 \times (-4) + 15 \div (-3)$

22 $5 \times (-3) + (-20) \div (-4)$



Step 4 Left to right

Medium

Questions 23–28

HOW TO

When the operations have the same rank, work from **left to right**.

$$-12 \div 3 \times 2 = -4 \times 2 = -8$$

$$(\text{not } -12 \div 6)$$

$$5 - 8 + 4 = -3 + 4 = 1$$

$$(\text{not } 5 - 12)$$

Evaluate. Work from left to right.

WE DO With your teacher: copy the working.

E1 $15 \div 3 \times (-5)$

E2 $-7 - 8 + 11$

YOU DO Your turn.

23 $-15 \div 3 \times (-2)$

$$= -5 \times (-2)$$

$$= 10$$

24 $-5 - 6 + 6$

$$= -11 + 6$$

$$= \underline{\quad}$$

25 $-20 \div 4 \times (-4)$

$$= \underline{\quad} \times (\underline{\quad})$$

$$= \underline{\quad}$$

26 $-9 - 10 + 2$

27 $6 \div (-2) \times (-2)$

28 $-7 - 10 + 8$

Step 5 Putting it all together

Challenging

Questions 29–32

HOW TO

One step a line. Copy down everything you have not worked out yet.

$$-2 - (3 + (-5)) \times 4$$

$$= -2 - (-2) \times 4$$

$$= -2 - (-8)$$

$$= 6$$

Evaluate. Brackets, then $\times$ and $\div$, then $+$ and $-$.

WE DO With your teacher: copy the working.

E1 $-3 - (-4 + 8) \times 3$

YOU DO Your turn.

29 $4 - (3 + 5) \times 3$

$$= 4 - 8 \times 3$$

$$= 4 - 24$$

$$= \underline{\quad}$$

30 $1 - (-7 + (-7)) \times (-2)$

$$= \underline{\quad} - (\underline{\quad}) \times (\underline{\quad})$$

$$= \underline{\quad} - \underline{\quad}$$

$$= \underline{\quad}$$

31 $4 - (5 + (-3)) \times 4$

32 $5 - (-3 + 6) \times 4$

Step 6 Is it true?

Challenging

Questions 33–36

HOW TO

Work out each side carefully, then compare.

Brackets change the answer:

$$(-4 + 6) \times 3 = 2 \times 3 = 6$$

$$-4 + 6 \times 3 = -4 + 18 = 14$$

Answer each question. Show your working.

WE DO With your teacher: copy the working.

E1 Put brackets in to make it true:
 $-6 + 3 \times 5 = -15$

E2 Mia says $-9 + 8 \times 5 = -5$. Is she right? Explain.

YOU DO Your turn.

33 Put brackets in to make it true:
 $-7 + 9 \times 3 = 6$

$$(-7 + 9) \times 3$$

$$= 2 \times 3$$

$$= \underline{\quad}$$

34 Mia says $-9 + 7 \times 3 = -6$. Is she right? Explain.

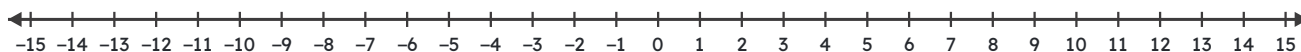
$$\underline{\quad} \times \underline{\quad} = \underline{\quad} \text{ first}$$

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

No: the answer is $\underline{\quad}$

35 Put brackets in to make it true:
 $-3 + 7 \times 4 = 16$

36 Mia says $-4 + 8 \times 2 = 8$. Is she right? Explain.



Step 1 The (−) key

Easy

Questions 1–10

HOW TO

Use the (−) key to enter a negative number. Use the − key to subtract. To work out $-35 + 12$, press:

(−) 3 5 + 1 2 =

Write the keys you press, then the answer.

WE DO With your teacher: copy the working.

E1 $-83 + 12$

YOU DO Your turn.

1 $-94 - 58$

2 $-37 - 41$

(−) $94 - 58 =$

(−) $37 - 41 =$

$= -152$

$=$

3 $-67 - 33$

4 $-25 - 83$

(−) $67 - 33 =$

$=$

Use your calculator.

WE DO With your teacher: copy the working.

E2 $-396 + (-140)$

E3 $-326 + 226$

YOU DO Your turn.

5 $-133 + (-216)$

-349

6 $148 + (-66)$

7 $-289 + (-210)$

8 $-395 + (-60)$

9 $127 + (-184)$

10 $-219 + (-194)$

Step 2 Multiplying and dividing

Easy

Questions 11–19

HOW TO

Enter each negative number with the (−) key. Some calculators need brackets around a negative number that comes after $\times$ or $\div$.

$-24 \times (-15) = 360$:

(−) 2 4 $\times$ ((−) 1 5) =

Use your calculator.

WE DO With your teacher: copy the working.

E1 -26×14

E2 $-533 \div (-41)$

YOU DO Your turn.

11 -36×41

$-1\,476$

12 $-160 \div 32$

13 $31 \times (-44)$

14 $-1\,215 \div 27$

15 -29×40

16 $-165 \div (-11)$

Write the keys you press, with brackets around the second number. Then write the answer.

WE DO With your teacher: copy the working.

E3 $880 \div (-22)$

YOU DO Your turn.

17 $-30 \times (-18)$

(−) $30 \times ((−) 18) =$

$= 540$

18 $-16 \times (-40)$

(−) $16 \times ((−) 40) =$

$=$

19 $392 \div (-14)$

$392 \div ((−) 14) =$

$=$

Step 3 Check the sign first

Medium

Questions 20–25

HOW TO

Before you press =, use the sign rules to decide if the answer is positive or negative. If the calculator disagrees, check how you entered the numbers. -48×25 : different signs, so negative. The answer is -1200 .

Write the sign of the answer first, then use your calculator.

WE DO With your teacher: copy the working.

E1 -34×29

YOU DO Your turn.

20 -43×58

sign: − (different signs)

$= -2\,494$

21 $-57 \times (-27)$

sign: + (same signs)

$=$

22 $-37 \times (-38)$

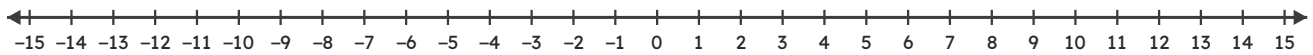
sign: +

$=$

23 $34 \times (-30)$

24 $-21 \times (-45)$

25 -39×46



Step 4 Brackets on the calculator

Medium

Questions 26–31

HOW TO

Enter brackets with the (and) keys. The calculator follows the order of operations.

$$(-35 - 47) \times 12 = -984:$$

(() (-) 3 5 - 4 7) × 1 2 =

Use your calculator. Write the value of the brackets first.

WE DO With your teacher: copy the working.

E1 $(-34 - 6) \times (-4)$

YOU DO Your turn.

26 $(-67 - 42) \times (-18)$

bracket: -109

$-109 \times (-18) = 1\,962$

27 $(-20 - (-31)) \times (-17)$

bracket: 11

$___ \times (___) = ___$

28 $(-70 - 61) \times 21$

bracket: $___$

$___ \times ___ = ___$

29 $(-55 - (-32)) \times (-14)$

30 $(-66 - 39) \times 11$

31 $(-87 - (-16)) \times 22$

Step 5 Estimate, then calculate

Challenging

Questions 32–37

HOW TO

Round each number to the nearest 10 and work it out in your head. Then use your calculator. The two answers should be close.

$$-48 \times 21 \approx -50 \times 20 = -1000$$

Calculator: -1008 . Close, so it makes sense.

Estimate first, then use your calculator.

WE DO With your teacher: copy the working.

E1 -33×15

YOU DO Your turn.

32 -38×13

estimate: $-40 \times 10 = -400$

calculator: $___$

33 -32×37

estimate: $___ \times ___ = ___$

calculator: $___$

34 -33×27

35 -61×17

36 -34×31

37 -83×35

Step 6 Calculator problems

Challenging

Questions 38–39

HOW TO

Write a number sentence first, then use your calculator.

Finish with a sentence that says what the answer means: a depth, a balance or a temperature.

Write a number sentence, use your calculator, then answer in words.

WE DO With your teacher: copy the working.

E1 An account has \$200. A fee of \$63 is taken out every month for 8 months. What is the balance now?

E2 A submarine at -465 m rises 33 m a minute for 10 minutes. How deep is it now?

E3 The lowest temperatures in Moscow for five days were -7°C , -20°C , -1°C , -9°C , -13°C . Find the average.

YOU DO Your turn.

38 An account has \$242. A fee of \$51 is taken out every month for 9 months. What is the balance now?

$242 - 51 \times 9$

$= -217$

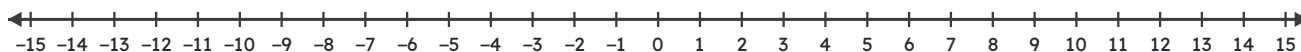
the balance is $___$

39 A submarine at -454 m rises 33 m a minute for 7 minutes. How deep is it now?

$___ + ___ \times ___$

$= ___$

it is at $___$ m



Step 1 Temperature

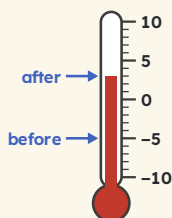
Easy

Questions 1–10

HOW TO

A rise is +, a fall is -. Start at the first temperature and count up or down the thermometer.

-5°C , then a rise of 8°C :
 $-5 + 8 = 3^{\circ}\text{C}$



Find the new temperature.
Write a number sentence in your head.

WE DO With your teacher: copy the working.

E1 -11°C , rises 16°C

E2 -4°C , falls 12°C

YOU DO Your turn.

1 1°C , falls 16°C

-15°C

2 12°C , falls 17°C

3 11°C , falls 17°C

4 -2°C , rises 14°C

5 -12°C , rises 6°C

6 12°C , falls 18°C

How many degrees did it rise? Count from the first to the second.

WE DO With your teacher: copy the working.

E3 from -6°C to 9°C

E4 from -5°C to 11°C

YOU DO Your turn.

7 from -10°C to -1°C

9°C

8 from -2°C to 1°C

9 from -11°C to 6°C

10 from -12°C to 9°C

Step 2 Above and below sea level

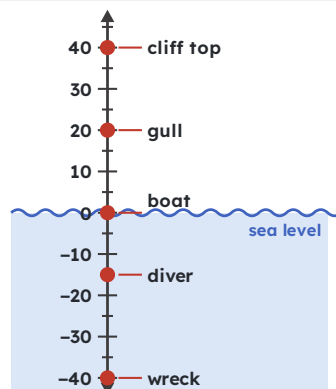
Easy

Questions 11–15

HOW TO

Heights above sea level are positive, depths below are negative.
Distance between two heights = higher - lower. When one is below sea level, add the two distances from sea level.

Use the diagram. How far apart are the two things?
Write a number sentence.



WE DO With your teacher: copy the working.

E1 the cliff top and the diver

YOU DO Your turn.

11 the gull and the wreck

$20 - (-40)$

$= 60 \text{ m}$

12 the cliff top and the boat

$40 - 0$

$= ___\text{ m}$

13 the diver and the wreck

$______ - (______)$

$= ___\text{ m}$

14 the gull and the diver

15 the cliff top and the wreck

Step 3 Lifts and floors

Medium

Questions 16–19

HOW TO

The ground floor is level 0. Basement levels are negative.
A lift at level -2 goes up 5 floors: $-2 + 5 = \text{level } 3$.



Write a number sentence, then answer the question.

WE DO With your teacher: copy the working.

E1 A lift is at level 3. It goes down 4 floors. Which level is it on now?

E2 How many floors does a lift travel from level -4 to level 4?

YOU DO Your turn.

16 A lift is at level 3. It goes down 7 floors. Which level is it on now?

$3 - 7$

$= \text{level } -4$

17 How many floors does a lift travel from level -3 to level 8?

$8 - (-3)$

$= 8 + 3$

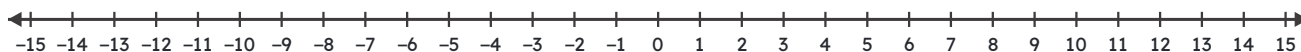
$= ___\text{ floors}$

18 A lift is at level -4 . It goes up 4 floors. Which level is it on now?

$______ + ______$

$= \text{level } ______$

19 How many floors does a lift travel from level -1 to level 3?

**Step 4 Money: credit and debt****Medium**

Questions 20–23

HOW TO

Money in (a deposit) is positive.
 Money out (a payment) is negative.
 A balance below zero means the account is **overdrawn**: you owe the bank.
 \$20, then pay \$35: $20 - 35 = -15$, so the balance is **-\$15**.

Ari's account starts with \$60.
 Find the balance after each transaction.

	Transaction	Amount
1	phone bill	-\$85
2	rent	-\$247
3	wages	+\$237
4	gift	+\$53
5	groceries	-\$75

WE DO

With your teacher: copy the working.

E1 Balance after transaction 1**YOU DO**

Your turn.

20 Balance after transaction 2

$$-25 - 247$$

$$= -\$272$$

21 Balance after transaction 3

$$-272 + 237$$

$$= \underline{\hspace{2cm}}$$

22 Balance after transaction 4

$$\underline{\hspace{2cm}} + \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}}$$

23 Balance after transaction 5**Step 5 Problems with several steps****Challenging**

Questions 24–25

HOW TO

Break the problem into steps. Write a number sentence for each step.
 Use $\times$ for a change that repeats.
 A diver at -40 m rises 3 m a minute for 5 minutes:
 $3 \times 5 = 15$
 $-40 + 15 = -25$ m

Write a number sentence for each step, then answer the question.

WE DO

With your teacher: copy the working.

E1 A diver is at -70 m. She rises 3 m a minute for 2 minutes. Where is she now?

E2 It is -6°C at 6 a.m. It warms 5°C an hour for 5 hours. What is the temperature now?

E3 A team scores $+2$ for a win and -1 for a loss. It wins 3 games and loses 6. What is its score?

YOU DO

Your turn.

24 A diver is at -40 m. She rises 3 m a minute for 4 minutes. Where is she now?

$$3 \times 4 = 12$$

$$-40 + 12$$

$$= \underline{\hspace{2cm}} \text{ m}$$

25 It is -3°C at 6 a.m. It warms 3°C an hour for 6 hours. What is the temperature now?

$$\underline{\hspace{2cm}} \times \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

$$\underline{\hspace{2cm}} + \underline{\hspace{2cm}}$$

$$= \underline{\hspace{2cm}} ^\circ\text{C}$$

Step 6 Quiz scores**Challenging**

Questions 26–28

HOW TO

In a quiz, a correct answer scores $+3$, a wrong answer scores -1 and no answer scores 0.

5 correct and 4 wrong:

$$5 \times 3 + 4 \times (-1) = 15 + (-4) = 11$$

Answer	Points
correct	$+3$
wrong	-1
no answer	0

Use the quiz scoring to answer each question.

WE DO

With your teacher: copy the working.

E1 Kai gets 2 correct and 5 wrong. What is Kai's score?

E2 Lea answers all 10 questions and scores 2. How many did she get correct?

YOU DO

Your turn.

26 Kai gets 2 correct and 3 wrong. What is Kai's score?

$$2 \times 3 = 6$$

$$3 \times (-1) = -3$$

$$\underline{\hspace{2cm}} + (\underline{\hspace{2cm}}) = \underline{\hspace{2cm}}$$

27 Lea answers all 12 questions and scores 28. How many did she get correct?

$$\text{try } \underline{\hspace{2cm}} : \underline{\hspace{2cm}} \times \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

$$\underline{\hspace{2cm}} \text{ wrong: } \underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

$$\underline{\hspace{2cm}} \text{ correct}$$

28 Kai gets 6 correct and 7 wrong. What is Kai's score?